

Erdős Problem #282

EP-282

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subseteq \mathbb{N}$ be an infinite set and consider the following greedy algorithm for a rational $x \in (0, 1)$: choose the minimal $n \in A$ such that $n \geq 1/x$ and repeat with x replaced by $x - \frac{1}{n}$. If this terminates after finitely many steps then this produces a representation of x as the sum of distinct unit fractions with denominators from A .

Does this process always terminate if x has odd denominator and A is the set of odd numbers? More generally, for which pairs x and A does this process terminate?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>